WEEK 09

Probability for data science

Mathematical foundations and Bayes classification



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A message with an unknown class

A new message arrives. We observe features, but its true label is hidden.

**“Special offer.
Follow this link.”**

Observed features
Offer = 1
Link = 1

P(spam | offer, link) = ?

The goal is to turn evidence into a probability, then choose a class.

Why probability belongs in data science

Data are finite and imperfect. New observations can have several plausible explanations.

Data-science task	How probability helps
Classify a message	Compare possible classes given its features
Interpret an unusual measurement	Judge how surprising it is under a model
Simulate future outcomes	Estimate how often an event could occur
Report uncertainty from a sample	Describe variation across possible samples

Every formula today helps describe uncertainty, combine evidence or support a decision.

Probability in the course workflow

We have described and cleaned observed data. Now we model uncertainty about outcomes and classes.

Earlier material	Today’s connection
Counts and rates	Events and their probabilities
Distributions and charts	A random variable’s distribution
Denominators and confounding	Conditional groups and base rates

One recurring question: which outcomes belong in the denominator?

Learning goals and lesson route

By the end, you should be able to explain each calculation in words.

Describe random outcomes with events and distributions.

Calculate conditional probabilities and combine evidence.

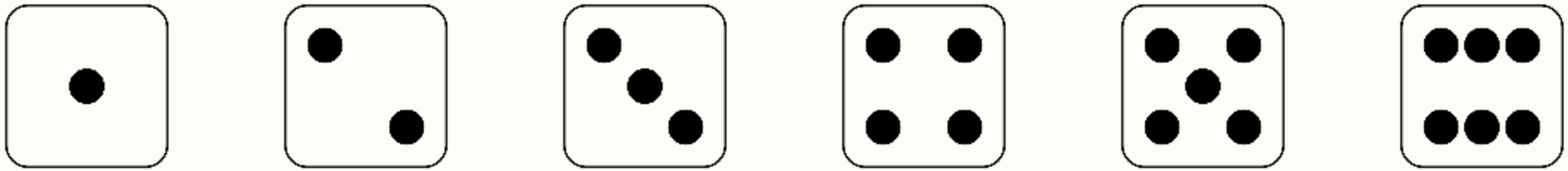
Estimate model probabilities through simulation.

Build a classifier from labelled data using Bayes' rule.

Two-hour route: foundations, conditional probability, Bayes and classification.

Outcomes, sample space and events

Experiment: roll one fair die and read the upper face.



Sample space Ω = {1, 2, 3, 4, 5, 6}

Outcome ω = 5
One result of the roll

Event A = {5, 6}
The result is greater than 4

What probability means mathematically

Probability assigns a consistent weight to each event under a model.

$$P(A) \in [0, 1]$$

P(A) is the probability that the outcome belongs to A.

0 means no probability mass. 1 means all probability mass.

Equally likely outcomes: $P(A) = |A| / |\Omega|$

The model specifies the possible outcomes and their weights.

Three axioms make probability consistent

These statements define a probability measure.

1. Non-negativity $P(A) \geq 0$

2. Total probability is one $P(\Omega) = 1$

3. Add disjoint events $P(A \cup B) = P(A) + P(B)$

The third rule applies when $A \cap B = \varnothing$ and extends to countably many events.

Useful rules follow from the axioms

A and its complement are disjoint and together fill the sample space.

$$\Omega = A \cup A^c$$

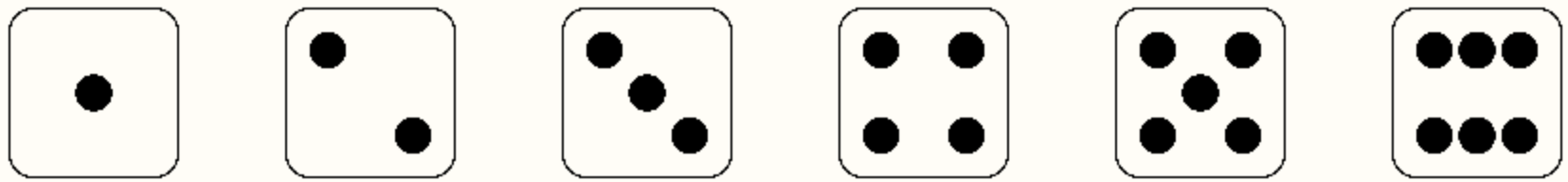
$$1 = P(\Omega) = P(A) + P(A^c)$$

$$P(A^c) = 1 - P(A)$$

Also: $P(\varnothing) = 0$, and $A \subseteq B$ implies $P(A) \leq P(B)$.

A die event: greater than four

All six outcomes are equally likely. A = {5, 6}.



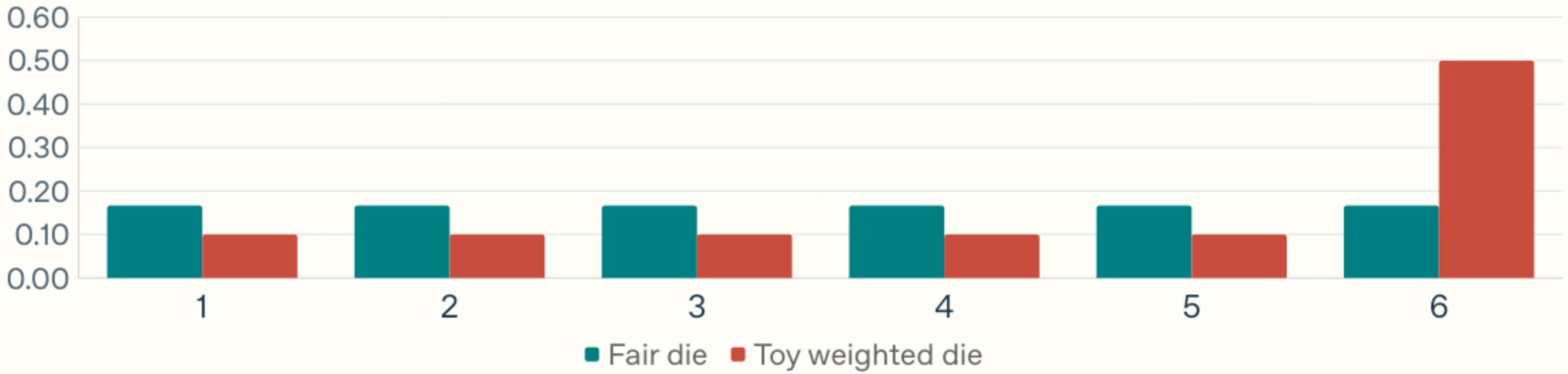
A: greater than 4					✓	✓
A^c: at most 4	✓	✓	✓	✓		

P(A) = 2/6 = 1/3

The highlighted outcomes supply the numerator. All six supply the denominator.

Counting works only for equally likely outcomes

A finite model can give different weights to different outcomes.

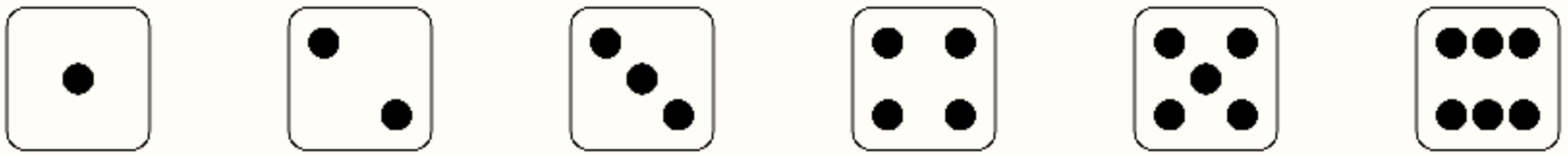


$$P(A) = \sum_{\omega \in A} P(\{\omega\})$$

For A = {5, 6}: fair model 1/3, weighted model 0.10 + 0.50 = 0.60.

Event operations on the same outcomes

A = {5, 6} and B = {4, 5, 6}. A is a subset of B.



A ∩ B: both					✓	✓
A ∪ B: either				✓	✓	✓
A^c: outside A	✓	✓	✓	✓		

$$A \subseteq B \Rightarrow P(A) \leq P(B)$$

An event is a set. “And”, “or” and “not” are set operations.

Complement and addition rules

An overlap must contribute once to the probability of “A or B”.

Complement $P(A^c) = 1 - P(A)$

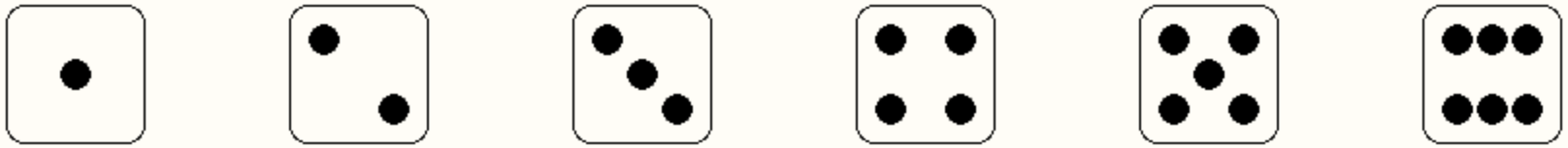
General addition $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Disjoint events $A \cap B = \varnothing \Rightarrow P(A \cup B) = P(A) + P(B)$

The subtraction corrects the double count of the shared outcomes.

Overlapping events: count the six once

E = “even” = {2, 4, 6}. H = “greater than 4” = {5, 6}.

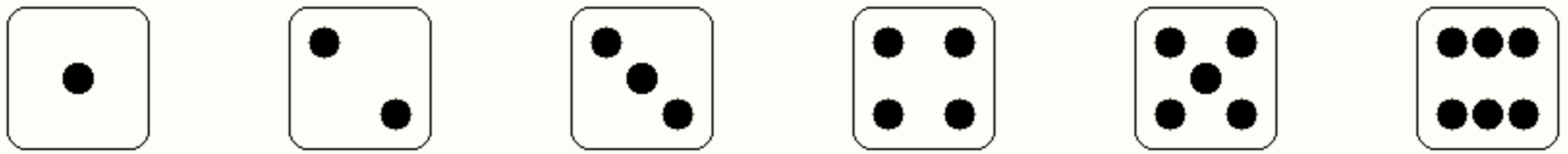


E: even		✓		✓		✓
H: greater than 4					✓	✓
E ∩ H						✓
E ∪ H		✓		✓	✓	✓

$$P(E \cup H) = 3/6 + 2/6 - 1/6 = 4/6$$

A random variable maps outcomes to numbers

X records the face value. Y records whether the result is greater than 4.



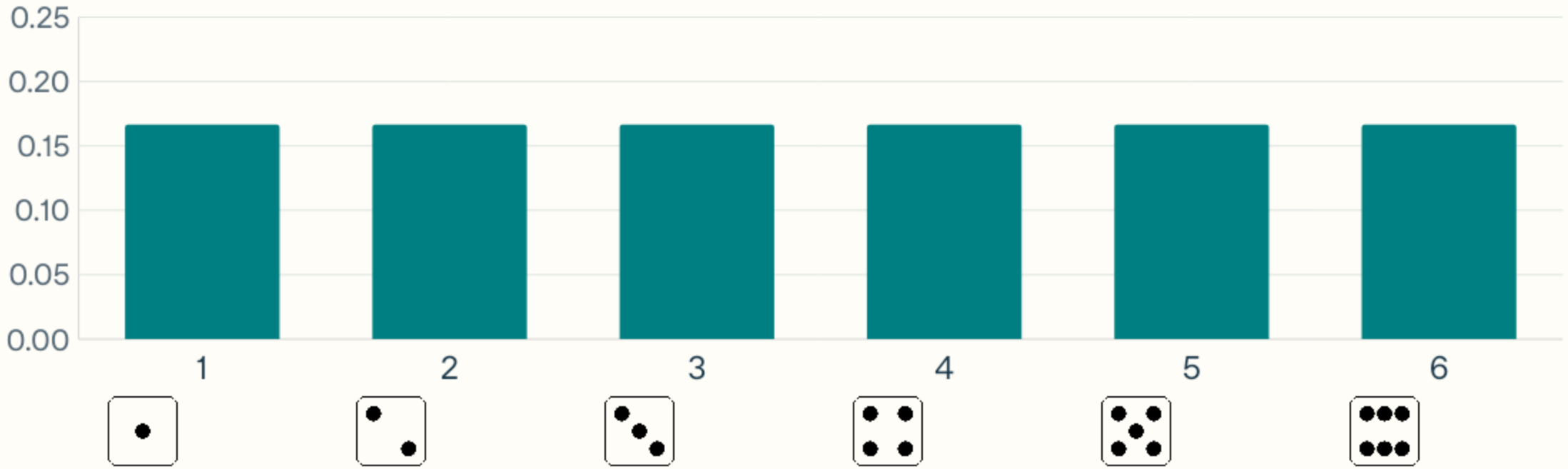
X(ω)	1	2	3	4	5	6
Y(ω)	0	0	0	0	1	1

$$X : \Omega \to \mathbb{R} \qquad Y = 1\{X > 4\}$$

The randomness comes from the outcome. The mapping itself is fixed.

The probability mass function

For a fair die, $p(x) = P(X = x) = 1/6$ for $x = 1, \ldots, 6$.



$p(x) \geq 0$ $\quad\quad$ $\sum_x p(x) = 1$

Expected value: a weighted average

The probability of each value supplies its weight.

$$E[X] = \sum_{x} x\, p(x)$$

$$= (1 + 2 + 3 + 4 + 5 + 6) / 6$$

$$= 3.5$$

A long-run average can lie between the possible results.

Variance: expected squared distance from the mean

Write $\mu = E[X]$. Squaring prevents positive and negative deviations from cancelling.

$$\mathrm{Var}(X) = E[(X - \mu)^2]$$

$$= E[X^2] - (E[X])^2$$

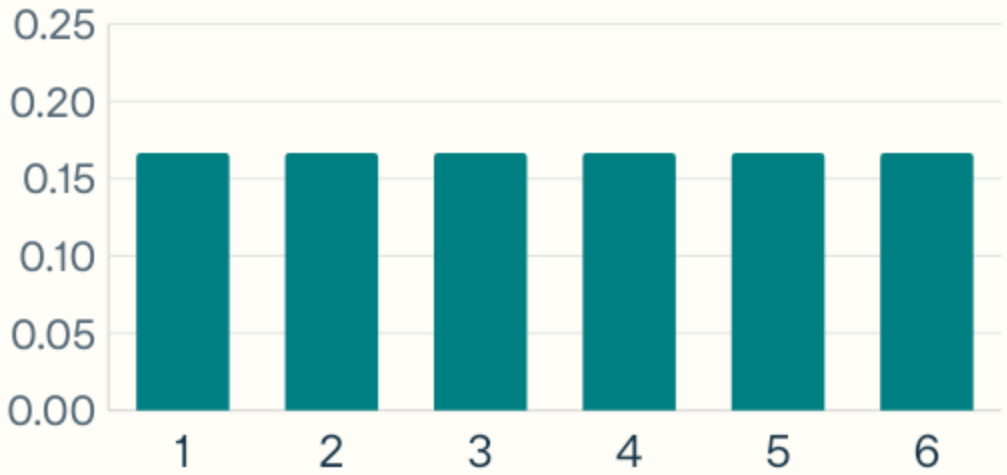
$$= 91/6 - 3.5^2 = 35/12 \approx 2.917$$

Standard deviation: $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)} \approx 1.708$.

Discrete mass and continuous density

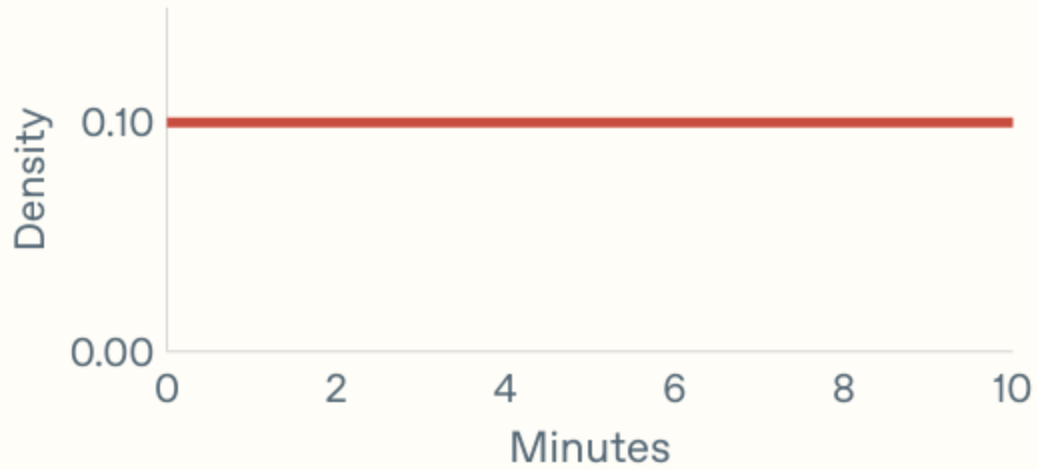
Counts use probability masses. Continuous measurements use areas under a density.

Discrete: die face



$P(X = 5) = 1/6$

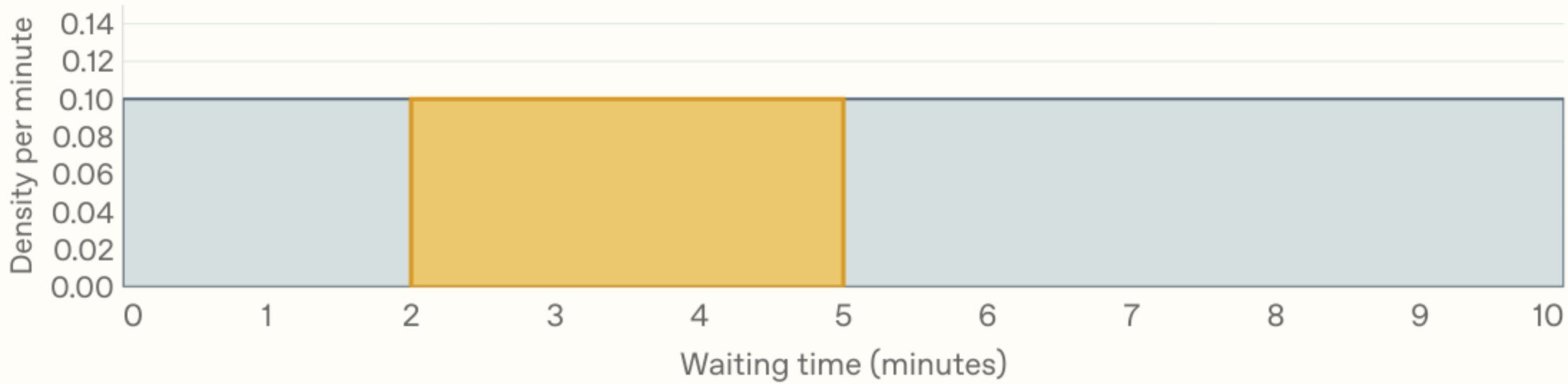
Continuous: waiting time



$P(2 \le T \le 5) = 0.30$

Probability is area under the density

T is uniform from 0 to 10 minutes. The highlighted interval is 2 to 5 minutes.



$$P(2 \le T \le 5) = \int_2^5 f(t)\,dt = 3 \times 0.1 = 0.30$$

Model probability and observed proportion

The model describes the experiment. The sample records one finite run.

Quantity	Calculation	What it describes
Model probability	$P(X > 4) = 1/3$	The fair-die model
Observed proportion	Success count / number of rolls	This particular run
One run with 10 rolls	$3 / 10 = 0.30$	An estimate of 1/3

The estimate may change between runs while the model probability stays fixed.

A reproducible die simulation

Each new call generates a fair roll. The seed lets us reproduce this teaching example.

```
import random
rng = random.Random(9)
rolls = [rng.randint(1, 6) for _ in range(10000)]

for n in [10, 100, 1000, 10000]:
    count = sum(x > 4 for x in rolls[:n])
    print(n, count, count / n)
```

True counts as 1 and False as 0. rolls[:n] keeps the first n rolls.

One simulated run

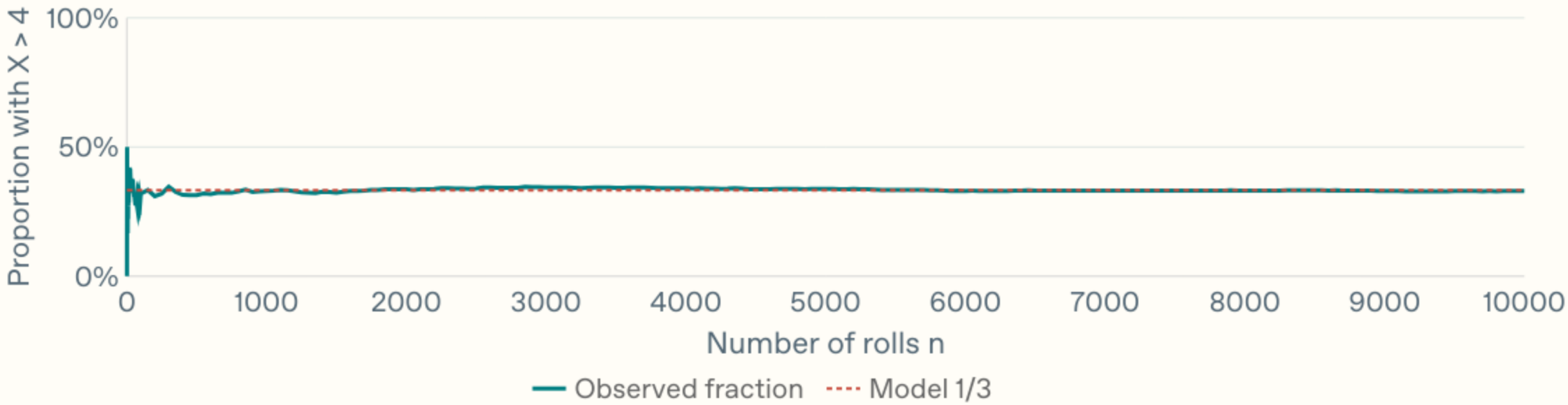
Python seed 9. All rows use prefixes of the same 10,000 stored rolls.

Number of rolls N	Count with X > 4	Count / N
10	3	0.3000
100	32	0.3200
1,000	330	0.3300
10,000	3,296	0.3296

The exact model value is 1/3 ≈ 0.3333. These counts come from the preceding code.

Relative frequency approaches model probability

The same 10,000 rolls as the Python example. Seed 9. Success means X > 4.

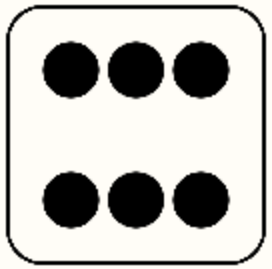


Convergence does not mean every extra roll moves the estimate closer.

Conditioning changes the reference group

Among rolls greater than 4, what fraction show 6?





Condition B = {5, 6}

$$P(X = 6 \mid X > 4)$$

$$= (1/6) / (2/6)$$

$$= 1/2$$

The denominator now describes B, because we know B occurred.

The direction of conditioning matters

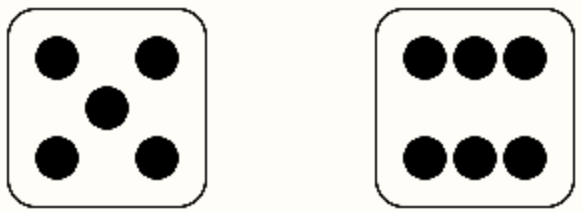
Lab 9 uses A = {5, 6} and B = {4, 5, 6}.

P(A | B)



2 matching outcomes / 3 in B
= 2/3

P(B | A)



2 matching outcomes / 2 in A
= 1

The conditional-probability formula

The same denominator reasoning works beyond equally likely outcomes.

$$P(A \mid B) = P(A \cap B) / P(B)$$

Part	Meaning
$P(A \cap B)$	Probability that both events occur
$P(B)$	Probability of the conditioning event, with $P(B) > 0$

For the die: (2/6) / (3/6) = 2/3.

The multiplication rule comes from conditioning

For $P(B) > 0$, start with the definition of conditional probability.

$$P(A \mid B) = P(A \cap B) / P(B)$$

Multiply both sides by $P(B)$.

$$P(A \cap B) = P(B)\, P(A \mid B)$$

If A and B are independent, this becomes $P(A \cap B) = P(A)P(B)$.

Joint, marginal and conditional probabilities

Counts in 1,000 fictional messages with known training labels.

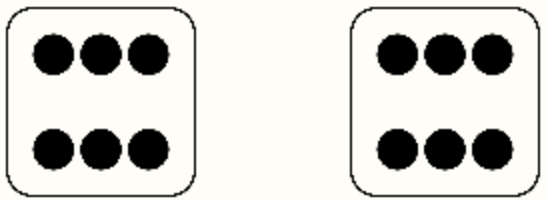
Class	Offer present	Offer absent	All messages
Spam	160	40	200
Regular	80	720	800
Total	240	760	1,000

Joint: P(spam ∩ offer) = 0.16. Marginal: P(spam) = 0.20.
Conditional: P(offer | spam) = 0.80.

Independence is different from disjointness

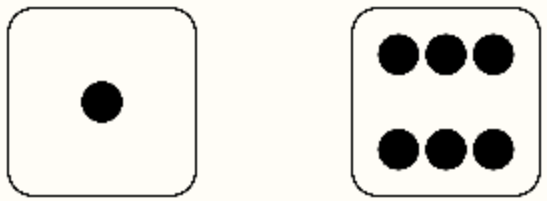
Independence describes probabilities. Disjointness describes shared outcomes.

Independent fair rolls



$P(\text{second} = 6 \mid \text{first} = 6) = 1/6$

Disjoint events on one roll



$P(X = 1 \text{ and } X = 6) = 0$
$P(X = 6 \mid X = 1) = 0$

Independent events satisfy $P(A \cap B) = P(A)P(B)$.

Checkpoint and ten-minute break

Take three minutes in pairs before the break. Use one fair die roll.

1. Let E mean “even” and H mean “greater than 3”.

2. Find $P(E)$, $P(H)$, $P(E \cap H)$ and $P(E \mid H)$.

3. Are E and H independent? Explain using probabilities.

After the break: a red ball is visible, but the chosen bag is hidden.

The two-bag experiment

Choose bag A or B with probability 1/2, then select one of its ten balls uniformly.



BAG A

8 red + 2 blue

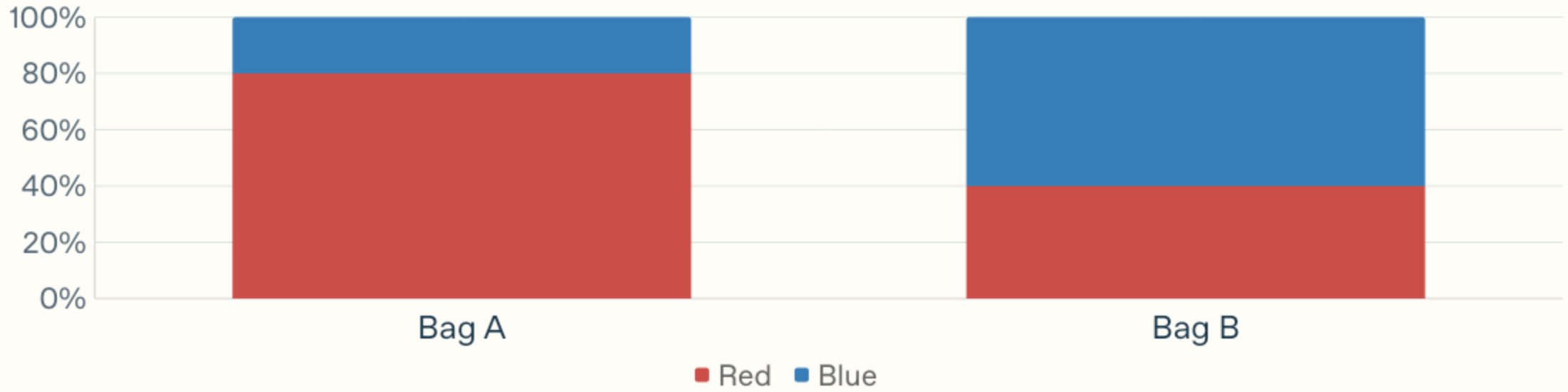


BAG B

4 red + 6 blue

The colour probabilities depend on the bag

Each bag supplies its own denominator of ten balls.



$P(R \mid A) = 8/10 = 0.80$

$P(R \mid B) = 4/10 = 0.40$

Expected outcomes in 1,000 trials

With equal bag selection, we expect 500 trials from each bag.

Chosen bag	Expected red	Expected blue	Expected total
A	500 × 0.80 = 400	500 × 0.20 = 100	500
B	500 × 0.40 = 200	500 × 0.60 = 300	500
Total	600	400	1,000

These are theoretical expected counts. A simulation can give different counts.

Probability paths through the experiment

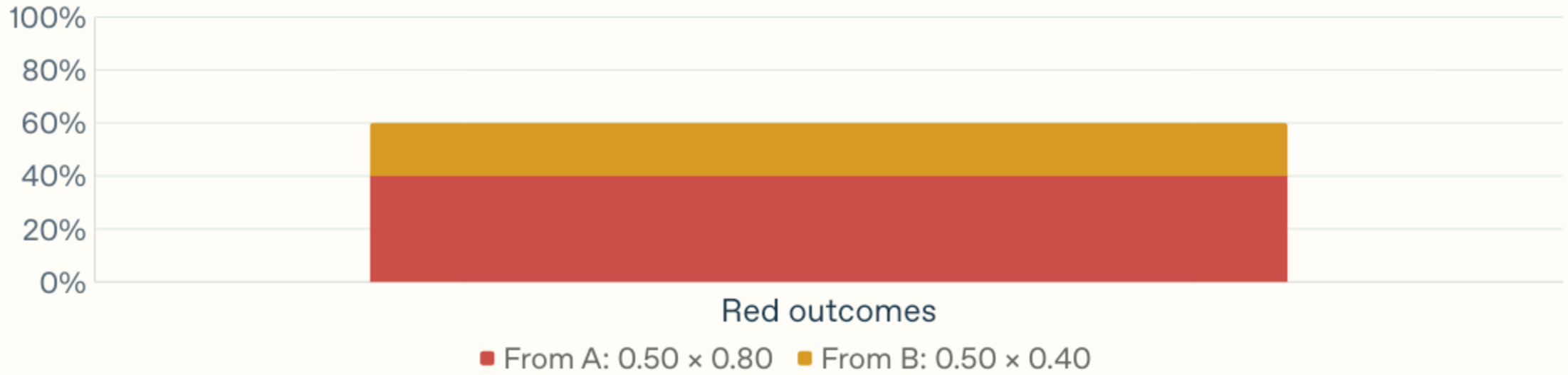
A complete path specifies both the chosen bag and the observed colour.

First event	Then	Multiply along the path	Joint probability
A: 0.50	Red: 0.80	0.50 × 0.80	0.40
A: 0.50	Blue: 0.20	0.50 × 0.20	0.10
B: 0.50	Red: 0.40	0.50 × 0.40	0.20
B: 0.50	Blue: 0.60	0.50 × 0.60	0.30

$$P(A \cap R) = P(A)\,P(R \mid A) = 0.40$$

Total probability adds the disjoint red paths

The red result can arrive through A or through B.



$$P(R) = 0.50 \times 0.80 + 0.50 \times 0.40$$

$$P(R) = 0.40 + 0.20 = 0.60$$

The law of total probability

Classes $H_1, \ldots, H_k$ form a partition: one and only one class occurs.

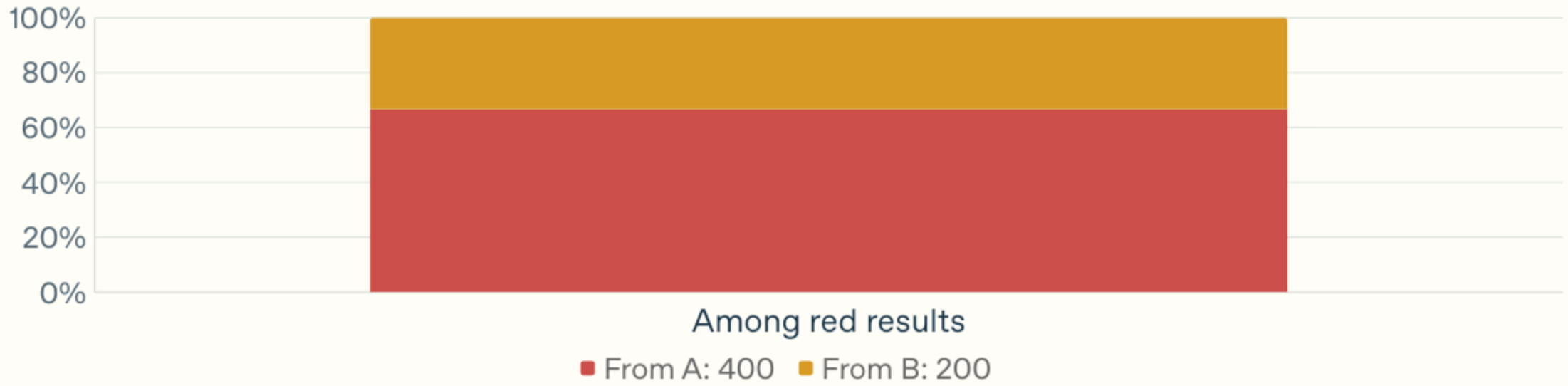
$$P(E) = \sum_i P(E \mid H_i)\, P(H_i)$$

Each class contributes its own path to E.
The contributions add because the classes are disjoint.

$$P(R) = P(R \mid A)P(A) + P(R \mid B)P(B)$$

Bayes reverses the question

After observing red, compare only the red routes.



P(A | R) = 400/(400 + 200) = 2/3

The four terms in Bayes' rule

Prior and likelihood form a joint probability. Evidence supplies the denominator.

$$P(A \mid R) = P(R \mid A)\, P(A) / P(R)$$

POSTERIOR
$P(A \mid R)$
Bag after red

LIKELIHOOD
$P(R \mid A)$
Red within bag A

PRIOR
$P(A)$
Bag before colour

EVIDENCE
$P(R)$
All red paths

Bayes’ rule follows by writing the joint two ways

Assume $P(E) > 0$ and $P(H) > 0$.

$$P(H \cap E) = P(H \mid E)P(E)$$

$$P(H \cap E) = P(E \mid H)P(H)$$

Set the right sides equal and divide by $P(E)$.

$$P(H \mid E) = P(E \mid H)P(H) / P(E)$$

Bayes' rule with every number shown

Equal priors: $P(A) = P(B) = 0.50$. Red probabilities: 0.80 and 0.40.

1. Numerator: $P(R \mid A)P(A) = 0.80 \times 0.50 = 0.40$.

2. Denominator: $P(R) = 0.40 + 0.20 = 0.60$.

3. Divide the matching joint probability by the evidence.

$$P(A \mid R) = 0.40 / 0.60 = 2/3$$

The same bags with a different prior

Now P(A) = 0.20 and P(B) = 0.80. The bag contents stay unchanged.

Chosen bag	Expected red	Expected blue	Expected total
A	200 × 0.80 = 160	200 × 0.20 = 40	200
B	800 × 0.40 = 320	800 × 0.60 = 480	800
Total	480	520	1,000

Predict the new P(A | R) before calculating it.

The posterior changes with the prior

Keep only the red outcomes in the new expected-count table.

P(A | R) = 160 / (160 + 320) = 1/3

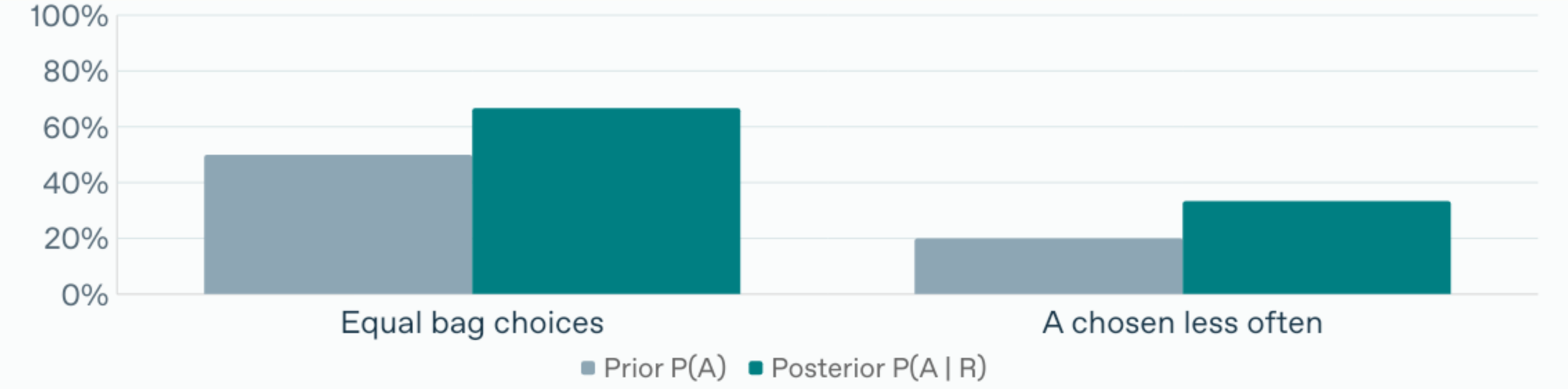
The new overall chance of red is P(R) = 480/1,000 = 0.48.

Seeing red raises the chance of A from 0.20 to about 0.333.

Red favours A, yet A remains less likely than B after observing red.

Same evidence, different starting probabilities

The bag contents stay fixed: P(R | A) = 0.80 and P(R | B) = 0.40.



Prior 50% gives posterior 66.7%. Prior 20% gives posterior 33.3%.

Classification predicts a categorical label

Let C be the class: spam or regular. Let X record the observed message features.

Stage	What is available	What the model does
Training	Features and known class labels	Estimate class and feature probabilities
Prediction	Features of a new message	Calculate P(C | X) for each class
Decision	Posterior probabilities and a rule	Return a predicted class label

The bag is now a class. The observed ball colour is now a message feature.

Training counts for a one-feature classifier

The same fictional training sample. Feature O is 1 when the word “offer” is present.

Known class	O = 1	O = 0	Total
Spam	160	40	200
Regular	80	720	800
Total	240	760	1,000

Learning the model begins with counting within each labelled class.

Estimated priors and likelihoods

Each fraction answers a different question about the training sample.

Model term	Calculation	Estimate
Prior P(C = spam)	200/1,000	0.20
Prior P(C = regular)	800/1,000	0.80
Likelihood P(O = 1 | spam)	160/200	0.80
Likelihood P(O = 1 | regular)	80/800	0.10

For prediction, we use these estimates as the probabilities in our fitted model.

Bayes classifies a message containing “offer”

First compute the probability of this evidence under the fitted one-feature model.

Spam route: 0.20 × 0.80 = 0.16.

Regular route: 0.80 × 0.10 = 0.08.

All offer messages: P(O = 1) = 0.16 + 0.08 = 0.24.

P(spam | O = 1) = 0.16/0.24 = 2/3

A posterior probability becomes a class decision

With equal error costs, choose the class with the larger posterior probability.

$$\hat{y} = \text{argmax } P(C = c \mid x), \text{ over classes } c$$

Candidate class	Posterior for an offer message	Decision
Spam	2/3 ≈ 0.667	Larger posterior
Regular	1/3 ≈ 0.333	Smaller posterior

This is the MAP rule: maximum a posteriori.

A second feature adds evidence

Feature L is 1 when the message contains a link. Counts are within each class.

Class	Class count	Offer present	Link present
Spam	200	160 (80%)	150 (75%)
Regular	800	80 (10%)	160 (20%)

```
prior = {'spam': 0.20, 'regular': 0.80}
theta = {'spam': (0.80, 0.75), 'regular': (0.10, 0.20)}
```

The Naive Bayes assumption

Given the class, the model treats the features as independent.

$$P(O, L \mid C) \approx P(O \mid C) \times P(L \mid C)$$

The condition “given the class” is essential.

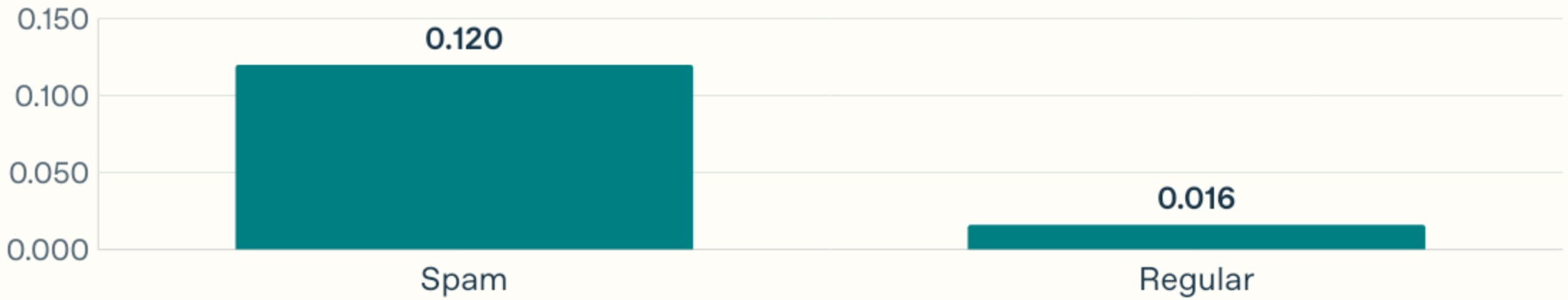
Real features can remain dependent even within a class.

Duplicating the same clue can make the model overconfident.

Bayes’ rule is exact. The feature-independence assumption simplifies the likelihood.

Class scores combine prior and feature evidence

The new message contains “offer” and a link. The model multiplies these factors.



Spam: 0.20 × 0.80 × 0.75 = 0.120

Regular: 0.80 × 0.10 × 0.20 = 0.016

Normalization gives class probabilities

Divide each class score by the total score for this same observed input.

P(spam | evidence) = 0.120/0.136 ≈ 0.8824



Regular: 0.016/0.136 ≈ 0.1176. The two probabilities sum to 1.

A small Naive Bayes classifier in Python

Use the prior and theta dictionaries estimated from the training counts.

```
def predict(offer, link):
    scores = {}
    for c in prior:
        a, b = theta[c]
        score = prior[c] * (a if offer else 1-a)
        scores[c] = score * (b if link else 1-b)
    z = sum(scores.values())
    return {c: s/z for c, s in scores.items()}
print(predict(1, 1))
```

Output: spam ≈ 0.8824, regular ≈ 0.1176. Try offer = 1, link = 0.

A classifier needs checks beyond this calculation

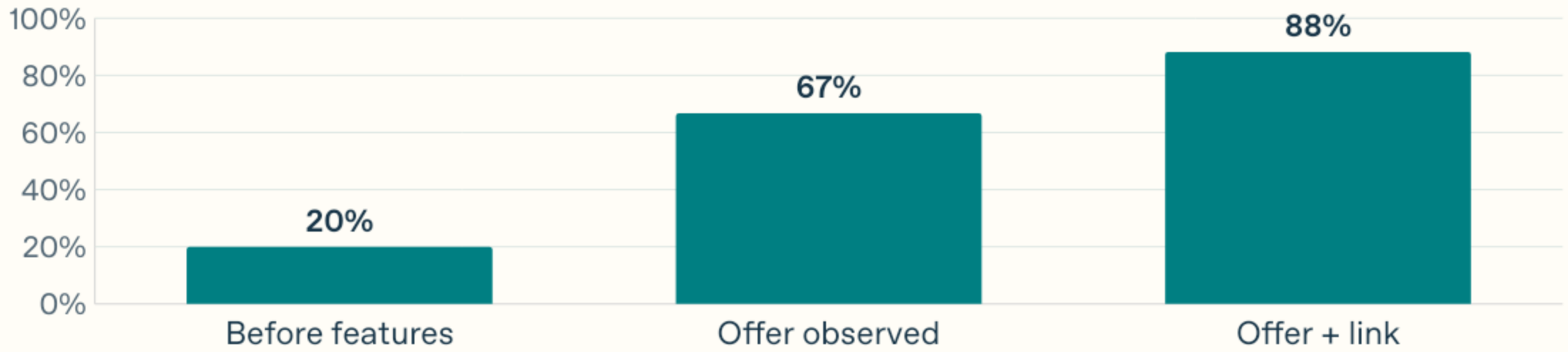
A correct Bayes calculation still depends on the data, assumptions and decision rule.

Check	What to do
Training and test separation	Fit probabilities on training data. Evaluate on unseen labelled data.
Feature dependence	Check whether several features repeat the same evidence.
Class frequencies	Check whether training priors suit the deployment population.
Decision consequences	Choose thresholds using validation data and the costs of mistakes.

Posterior probabilities describe the fitted model. Calibration checks their reliability.

Evidence updates the spam probability

The synthetic model answers the question from the opening slide.



With both features, MAP predicts spam with a posterior of about 88.2%.

The core probability relationships

The purpose is to identify events, reference groups and evidence before calculating.

Relationship	Calculation
Complement	$P(A^c) = 1 - P(A)$
Conditional probability	$P(A \mid B) = P(A \cap B) / P(B)$, with $P(B) > 0$
Joint probability	$P(A \cap B) = P(A \mid B)P(B)$
Bayes classification score	Score(class) = prior × likelihood
Posterior normalization	P(class | evidence) = score / sum of class scores

Naive Bayes models the feature likelihood as a product of probabilities given the class.

Exit questions and next week

Explain each answer using its reference group.

How do model probabilities differ from proportions in one sample?

How does Bayes turn feature likelihoods into class probabilities?

What assumption allows Naive Bayes to combine several features?

Next: sampling variability and inference. Later: classification and evaluation.

Reading: OpenIntro Ch. 3. ISLP Ch. 4. Reference slides follow this lesson.

Expected counts in repeated trials

An expected count is the long-run average count under the model.

Expected count = N × P(event)

For 10,000 rolls and X > 4: 10,000 × 1/3 ≈ 3,333.3.

For one face: 10,000 × 1/6 ≈ 1,666.7.

An expected count can be non-integer. Every observed count is an integer.

Reversing the condition changes the question

Both expressions describe the same overlap, viewed within different groups.

P(A | B): among 4, 5 and 6, how often is the result 5 or 6?

P(B | A): among 5 and 6, how often is the result at least 4?

$P(A \mid B) = 2/3$ $P(B \mid A) = 1$

A conditional probability usually changes when we reverse the condition.

Prior, likelihood and posterior

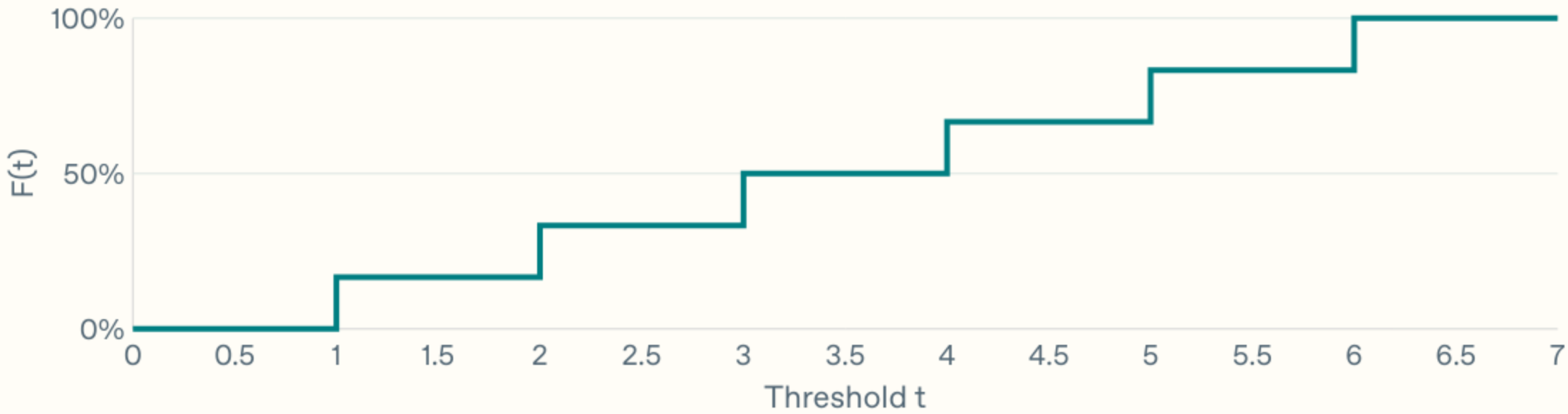
These numbers describe different questions, even when they concern the same experiment.

Quantity	Value	Interpretation
Prior P(A)	0.50	Before seeing colour, half the trials choose A
Likelihood P(R | A)	0.80	Within A, eight of ten balls are red
Posterior P(A | R)	2/3 ≈ 0.667	Among red trials, about two thirds choose A

Seeing red raises the chance of A from 50% to about 66.7%, under this model.

The cumulative distribution function

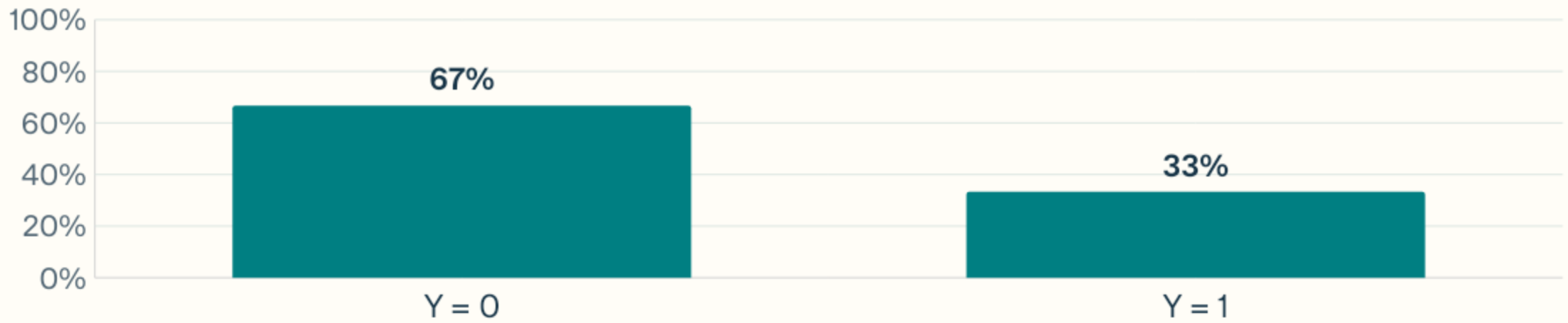
$F(t) = P(X \le t)$. For a fair die, the CDF increases in steps of 1/6.



For example, $F(4) = P(X \le 4) = 4/6$.

Bernoulli: one event becomes zero or one

Y = 1 if the die shows more than 4, and Y = 0 otherwise.

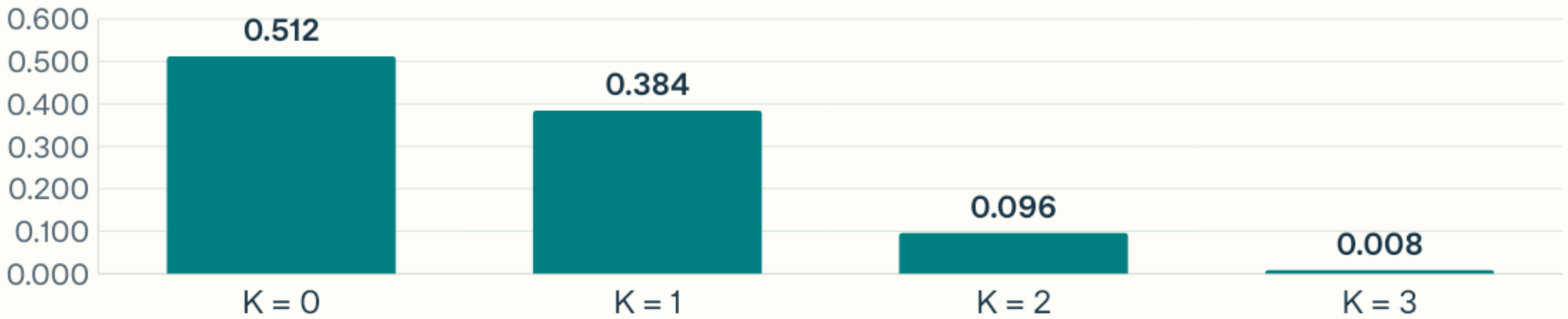


P(Y = 1) = p = 1/3

E[Y] = p. Var(Y) = p(1 – p) = 2/9.

Binomial: count successes in n independent trials

Example: n = 3 and p = 0.20. K counts successes.

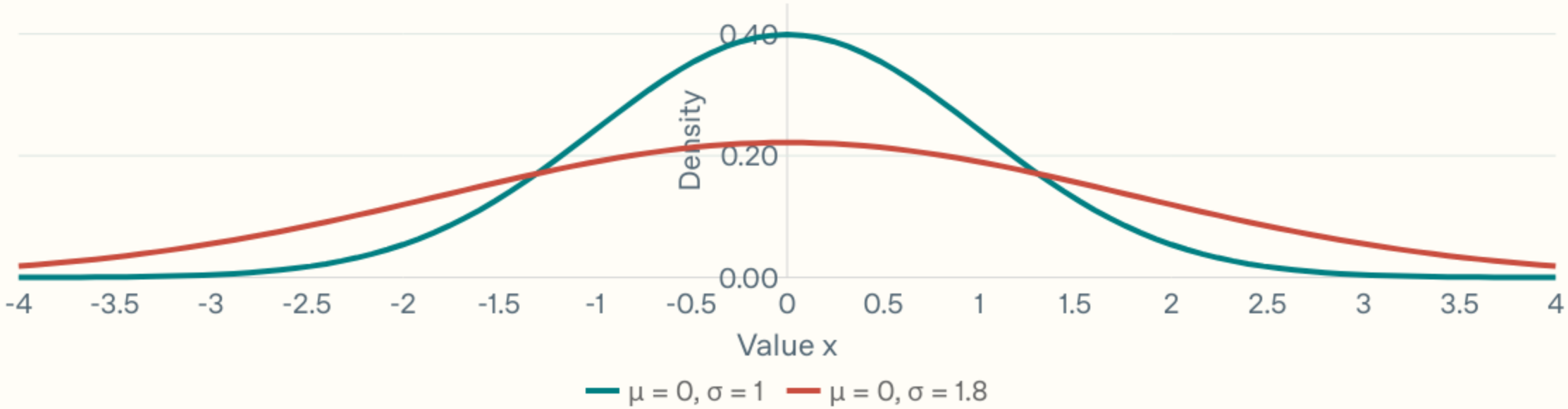


$$P(K = k) = C(n,k)\, p^k\, (1 - p)^{n-k}$$

$$P(K = 2) = 3 \times 0.20^2 \times 0.80 = 0.096.$$

Normal density: location and spread

A normal model has mean μ and standard deviation σ > 0.



Larger σ spreads the density. Probabilities remain areas under the curve.

Continuous features in Naive Bayes

Toy measurement x = 2. Use one normal likelihood per class and equal class priors.

Class	Mean μ	SD σ	Density at x = 2
A	0	1	f(2 | A) ≈ 0.05399
B	3	1	f(2 | B) ≈ 0.24197

$$P(B \mid x = 2) \approx 0.24197/(0.05399 + 0.24197) \approx 0.818$$

The feature likelihoods are densities. The class posterior is a probability.

Bayes classification with several classes

For one fixed input, calculate a score for every mutually exclusive class.

Class	Prior	Likelihood	Score	Posterior
A	0.40	0.20	0.080	0.40
B	0.35	0.20	0.070	0.35
C	0.25	0.20	0.050	0.25

Total score = 0.20. MAP chooses class A.

The winning class can have a posterior below 50%.

Smoothing avoids zero estimates

For one binary feature, k of n training examples in a class have value 1.

Smoothed p = (k + α) / (n + 2α)

Laplace smoothing uses α = 1.

If k = 0 and n = 200: p = 1/202 ≈ 0.00495.

For feature value 0, use the complementary probability 1 – p.

No observed examples does not establish that an event is impossible.

Log scores for many features

Multiplying many small positive likelihoods can cause numerical underflow.

$$\log \text{score}(c) = \log P(c) + \Sigma_j \log P(x_j \mid c)$$

$\log(0.120) \approx -2.1203$. $\log(0.016) \approx -4.1352$.

Logarithms preserve the ordering, so MAP still selects spam.

Stable normalization: subtract the largest log score before exponentiating.

A factory alarm and a rare defect

A constructed model for 10,000 items. These figures are teaching assumptions.

1% of items have a defect.

The alarm activates for 90% of defective items.

The alarm also activates for 5% of good items.

If an alarm activates, what is the chance that the item is defective?

Most alarms can come from good items

Expected counts under the constructed factory model.

True state	Alarm	No alarm	All items
Defective	90	10	100
Good	495	9,405	9,900
Total	585	9,415	10,000

$P(\text{defect} \mid \text{alarm}) = 90/585 \approx 15.4\%$

Simulating the two-bag experiment

A fresh random draw selects the bag, and another fresh draw selects the colour.

```
import random
rng = random.Random(9)
p_a = 0.50
records = []
for _ in range(10000):
    bag = 'A' if rng.random() < p_a else 'B'
    p_red = 0.80 if bag == 'A' else 0.40
    red = rng.random() < p_red
    records.append((bag, red))
```

For Lab 9 Part 7, change p_a to 0.20 and run the whole experiment again.

Estimating Bayes' result from the records

For P(A | R), the reference group contains only records where red is True.

```
n_red = sum(red for bag, red in records)
n_a_red = sum(bag == 'A' and red for bag, red in records)
posterior = n_a_red / n_red if n_red else None
print(n_a_red, n_red, posterior)
```

Estimate = A-and-red count / all-red count

Compare with the exact result: 2/3 for p_a = 0.50, or 1/3 for p_a = 0.20.

Common errors and their corrections

A full sentence often reveals the denominator mistake.

Error	Correction
$P(A \mid R) = P(R \mid A)$	Read each expression as “among …”
Use only A’s red route in $P(R)$	Include red outcomes from every possible bag
Treat an expected count as guaranteed	Separate the model expectation from one observed run
Reuse one random number for both stages	Make a fresh random draw for the colour

Check the experiment, then the numerator, then the denominator.

Pair exercise: a new selection mechanism

Same reasoning, new numbers. All values below define a hypothetical model.

Model probability	Value
P(A) and P(B)	0.30 and 0.70
P(R | A) and P(R | B)	0.60 and 0.20

1. Calculate expected red counts from each bag in 1,000 trials.

2. Find P(R), then P(A | R). Explain both denominators.

Write the count argument before substituting into Bayes' rule.

Worked solution to the pair exercise

The expected red counts are 180 from A and 140 from B.

Step	Calculation
A and red	1,000 × 0.30 × 0.60 = 180
B and red	1,000 × 0.70 × 0.20 = 140
Overall red probability	(180 + 140) / 1,000 = 0.32
A given red	180 / 320 = 0.5625 = 56.25%

The prior for A was 30%. After red, its posterior is 56.25%.